

Proof-of-Concept Summary

A working reference implementation — not a simulation — of NIST-standardized post-quantum cryptography combined with homomorphic encryption for fintech

This proof-of-concept demonstrates a stack that could plausibly sit inside a fintech’s own infrastructure as it migrates to comply with Nigeria’s CBN data localization mandate (deadline: 1 January 2027) — the same re-architecture moment being the cheapest time to also build in quantum-safety and privacy-preserving analytics.

SIGNATURE STANDARD	KEY EXCHANGE STANDARD	PRIVACY LAYER	TEST COVERAGE
ML-DSA-65	ML-KEM-768	CKKS (HE)	18 / 18 passing

What This Demonstrates

Capability	Primitive	Real-world use case
Quantum-safe identity credentials	ML-DSA-65 (FIPS 204)	A bank/KYC unit issues a signed credential; any relying party verifies it offline, no callback to the issuer
Quantum-safe channel security	ML-KEM-768 (FIPS 203)	A shared secret between a fintech backend and a Nigerian data-centre endpoint, resistant to “harvest now, decrypt later”
Privacy-preserving payment analytics	CKKS homomorphic encryption	A credit/fraud scoring engine computes a sum, mean, and threshold over encrypted transaction data — and never sees a raw amount

Why This Matters Now

Financial data has a long shelf life — KYC records and transaction histories are often retained for years. Data encrypted with classical algorithms today can be captured today and decrypted later once quantum computers are capable (“harvest now, decrypt later”), making PQC migration a present-tense risk for financial records, not a distant one.

At the same time, open banking and data-sharing initiatives need a way for institutions to compute over each other’s data — for credit scoring, fraud detection, cross-institution verification — without exposing raw customer records. Homomorphic encryption is one of the few real, working answers to that problem.

Engineering Honesty: Real Trade-offs

- **ML-DSA-65 signatures are larger** (~3.3 KB) than classical ECDSA — a real, known cost of PQC signatures worth planning for in bandwidth/storage-constrained systems.
- **ML-KEM-768 is close to a drop-in replacement** for classical key exchange (~1.1 KB ciphertext overhead) in most protocols.
- **CKKS is an approximate scheme**, suited to real-valued aggregates (sums, means, thresholds) — not appropriate for computations requiring exact integer precision without additional care around scale management.

- **This demo keeps key material in process memory** for simplicity. A production deployment would keep the ML-DSA issuer key and the CKKS secret key in an HSM or managed KMS.

How to See It Running

- **Scripted walkthrough, no server needed:** ``python demo.py``
- **Live API with interactive docs:** ``uvicorn app.main:app --reload --port 8000``, then open ``/docs``
- **Portable via Docker:** ``docker build -t qsafe-fintech .`` then ``docker run -p 8000:8000 --env-file .env qsafe-fintech``

Relationship to CryptoRadar

CryptoRadar (open-source, github.com/bennybriel/cryptoradar) diagnoses where an existing fintech codebase relies on quantum-vulnerable cryptography — hardcoded secrets, RSA/ECDSA usage, weak hashes. This proof-of-concept is the reference implementation of the migration path those findings point toward: what the fix actually looks like once built, running as real, tested code rather than a proposal.

Engagement

If you're a fintech, bank, or payment processor evaluating post-quantum migration or CBN data localization compliance, this repository is the working proof-of-capability behind a Quantum-Safe & Data-Sovereignty Readiness Assessment.

Contact: **bennybriel@gmail.com**